

1. Calibrate without counters
Small random ring: $A \rightarrow B \rightarrow C \rightarrow A$
Large random ring: $A \rightarrow \dots \rightarrow A$
1M single-load timings per point

2. Freeze a timing threshold
Fit hit/miss distributions
Check with independent seeds
No PMU inputs to classifier

3. Held-out workloads
Change footprint and node reuse
Time one load at a time
3 repeats $\times$ 1M accesses/case

Actual sample: fence $\rightarrow$ timer $\rightarrow$ load $\times p \rightarrow$ timer
Save raw delta; optionally repeat p , otherwise $p = \text{next}$
Count hit when $\text{delta} \leq \text{frozen threshold}$

4. Validation copy: same exact assembly loop
PMU counts only around 1M-load loop
Reference = $1 - \text{L1 misses} / \text{retired loads}$
Check auxiliary loads; compare error and uncertainty